

Top 50 Real-Time Networking Interview Questions with In-Depth Answers

1. What happens when you type google.com in a browser?

DNS resolves the domain name to an IP address, ARP resolves the gateway MAC address, TCP handshake is established, and then the browser sends an HTTPS request to retrieve the webpage.

2. User says “Internet is not working.” How do you troubleshoot?

Check IP configuration, ping localhost, ping own IP, ping gateway, ping 8.8.8.8, and ping google.com to isolate whether the issue is local, gateway, internet, or DNS related.

3. How do you identify whether the issue is DNS or internet?

Ping an external IP like 8.8.8.8. If it works but domain names fail, the issue is DNS.

4. A PC got APIPA address (169.254.x.x). What does it mean?

It means DHCP failed, so the system assigned itself an automatic private IP address.

5. Why can two PCs in same VLAN not communicate?

Possible causes are wrong subnet mask, wrong VLAN assignment, duplicate IP, port security, or firewall issues.

6. One switch port is down. How do you troubleshoot?

Check the cable, verify interface status, confirm the port is not shutdown, and verify speed/duplex settings.

7. What is MAC flapping?

It is when the same MAC address appears on multiple switch ports, often due to loops.

8. How does STP prevent loops?

STP blocks redundant paths by electing root and designated ports while blocking others.

9. What is a broadcast storm?

An excessive amount of broadcast traffic causing network congestion, usually due to switching loops.

10. How do you find VLAN mismatch?

Use show interfaces trunk and show vlan brief to confirm VLAN assignments and trunk allowed VLANs.

11. Why can hosts in different VLANs not communicate?

They require inter-VLAN routing using a router or Layer 3 switch.

12. What happens if default gateway is wrong?

The host can communicate locally but cannot reach remote networks.

13. Static route is configured but traffic fails. Why?

Possible reasons are wrong next hop, missing return route, interface down, or ACL issues.

14. What is asymmetric routing?

Traffic takes one path to destination and a different path back, causing issues with stateful devices.

15. Why is OSPF neighbor stuck in INIT?

Hello packets are received but bidirectional communication is not established.

16. DHCP is not assigning IPs. What do you check?

Check DHCP pool, VLAN configuration, DHCP relay, and scope exhaustion.

17. Why use ip helper-address?

It forwards DHCP broadcasts across routers to the DHCP server.

18. DNS server reachable but names not resolving. Why?

Possible issues are wrong DNS records, port 53 blocked, or zone problems.

19. Difference between DHCP Discover and Offer?

Discover is sent by the client requesting an IP, Offer is sent by the server proposing an IP.

20. Why does DNS use both TCP and UDP?

UDP is used for standard queries, TCP for zone transfers and large responses.

21. OSPF neighbors not forming. Why?

Check area mismatch, timers, authentication, and subnet configuration.

22. EIGRP neighbors down. Why?

Check AS number mismatch, K-values mismatch, or interface issues.

23. OSPF DR election depends on what?

Interface priority first, then router ID.

24. Why is route missing in routing table?

Route may not be advertised, may be filtered, or neighbor may be down.

25. Administrative distance meaning?

It determines which routing source is preferred; lower values are preferred.

26. NAT configured but internet not working. Why?

Check ACL, inside/outside interfaces, and default route.

27. How do you verify NAT translations?

Use show ip nat translations.

28. ACL configured but traffic still allowed. Why?

ACL may be applied in the wrong direction or on the wrong interface.

29. Difference between inbound and outbound ACL?

Inbound filters packets entering an interface; outbound filters leaving packets.

30. Why does ACL order matter?

ACLs are processed top to bottom; first match wins.

31. How do you check routing table?

Use show ip route.

32. How do you verify interface status?

Use show ip interface brief.

33. How do you trace packet path?

Use traceroute to see each hop.

34. How do you verify ARP entries?

Use show arp.

35. How to see MAC address table?

Use show mac address-table.

36. What is Port Security?

It restricts MAC addresses allowed on a switch port.

37. What is BPDU Guard?

It shuts down a PortFast port if BPDUs are received.

38. What is DHCP Snooping?

It blocks unauthorized DHCP servers.

39. What is DAI?

Dynamic ARP Inspection prevents ARP spoofing.

40. Why disable unused ports?

To prevent unauthorized access.

41. Users in one VLAN cannot access server VLAN. What do you check?

Check inter-VLAN routing, ACLs, and gateway configuration.

42. Ping works but application fails. Why?

The network path is working but the application port or service may be blocked.

43. Website slow but reachable. Why?

Possible causes are high latency, bandwidth congestion, or DNS delays.

44. Duplicate IP symptoms?

Intermittent connectivity and ARP conflicts.

45. One-way ping possible?

Yes, due to ACLs or missing return routes.

46. Difference between collision domain and broadcast domain?

A switch port is one collision domain; each VLAN is one broadcast domain.

47. Why full duplex removes collisions?

Because sending and receiving happen simultaneously.

48. Why use Layer 3 switch instead of router?

Layer 3 switches route faster using hardware ASICs.

49. What is MTU mismatch?

Different packet size settings can cause fragmentation or drops.

50. What is the first thing you check during troubleshooting?

Determine the scope: one user, one VLAN, or the whole network.